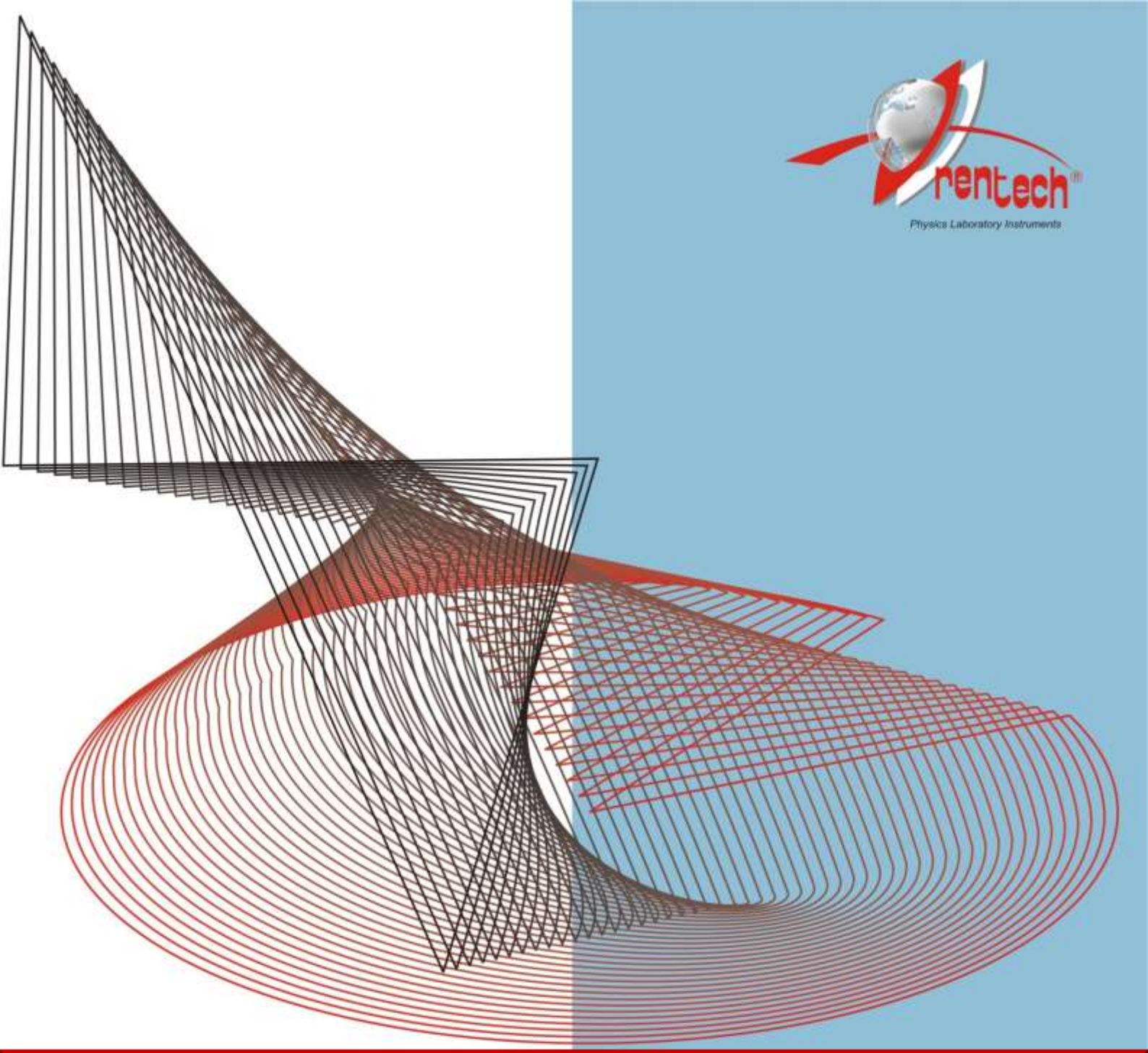


MEASUREMENTS, ERRORS AND GRAPHS

(Economic Version)

Student's Guide

Laboratory Manual and Workbook

Ankara-2015

Package List

1. A Set of Objects with Different Shapes
2. Digital Balance
3. Micrometer
4. Caliper
5. Graduated Cylinder
6. Ruler
7. A Set of Mass with Different Weights
8. Instructional Manuals (**Teacher and Student Guides**)

Note: This version of the product does NOT include the spherometer, the set of lenses with different focal points and the equal-arm balance unless specifically requested in the purchase order.They are included in the advanced version of the product.

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1. Purpose

The purpose of this experiment is to;

1. Learn how to use caliper and micrometer,
2. Measure external dimensions (length, volume) of a given regular body,
3. Determine depth of a given object using a caliper,
4. Determine the possible **errors** in the measurements by ruler, caliper and micrometer, and compare their **accuracy**,
5. Determine the density of a regular body of known mass using different measurement techniques.

2. Physics and Measurement

Physics is one of the most fundamental sciences. It deals with behavior and structure of matter. It is based on experimental observations and quantitative measurements. In order to identify meaningful relationships that characterize nature, it is important to know how to measure physical quantities in terms of which the laws of physics are stated.

Physical quantities are not generally independent to one another. Physics is composed of seven fundamental quantities. In mechanics, there are three fundamental quantities which are length, mass and time. The other quantities used in mechanics can be derived in terms of these three. The other four are; current, thermodynamic temperature, amount of substance and luminous intensity.

2.1. International System of Units (SI Units)

To make accurate and reliable measurements, a standard must be defined that do not change in various locations. Scientists and engineers all over the world use the same system of units as a standard to measure physical quantities.

This system is the International System of Units (SI Units). It is based on the metric system and basic set of units including meter, kilogram and second. The SI base units are given in the Table-(1) below.

Table-**1**: Fundamental quantities and their units

Base Quantitiy	SI Base Unit	Symbol
Length	**meter**	*m*
Mass	**kilogram**	kg
Time	**second**	s
Current	**ampere**	A
Thermodynamic Temperature	**kelvin**	K
Amount of Substance	**mole**	mol
Luminous Intensity	**candela**	cd

2.2. The Prefixes

In metric system, the larger and smaller units are defined in multiplies of 10 from the standard unit. Commonly used prefixes are listed in Table-(2).

Table-**2**: Commonly used prefixes

Power of ten	Prefix	Symbol
10^{-15}	femto	**f**
10^{-12}	pico	**p**
10^{-9}	nano	**n**
10^{-6}	micro	**μ**
10^{-3}	milli	**m**
10^{-2}	centi	**c**
10^{3}	kilo	**k**
10^{6}	mega	**M**
10^{9}	giga	**G**
10^{12}	tera	**T**

2.3. Types of Measurements

In general, measurements are divided into two groups, namely direct and indirect measurement. Direct measurements of a physical variable are obtained by the reading of the scale of an instrument calibrated in the corresponding units of the variable.

Example of a direct measurement is that measuring a certain length with a ruler. Indirect or calculated measurements are the deduction of the magnitude of physical variables using mathematical operations with magnitudes obtained by direct measurements.

In general terms, indirect measurements are those that correspond to physical variables whose values have not been determined by an instrument that can measure them directly. As an example, calculation of a sphere (of which radius,r, is measured initially) by the formula of $V = \frac{4}{3}\pi r^3$.

3. Errors

No physical quantity can be measured with a perfect certainty; there are always errors in a measurement. It is simply an observed fact that when independent measurements are repeated on a physical system, all the measured values will not generally be exactly the same. In general, the types of errors in a particular experiment may be due to the observer, experimental equipment or the design of the experiment. In recording a measurement, it is important to estimate and record the error. Error in a measurement can be indicated by making a reasonable assessment of the accuracy of the measurement in terms of the sources of errors and estimating the maximum possible error in the measurement. Then, the result of the measurement is written by the maximum possible error using the symbol "$\pm$" followed from the value observed from the scale. As an example, consider "x" is the actual measurement and "Δx" is the maximum reading error, then the result of the measurement is written as "$x \pm \Delta x$".

3.1. Types of Errors

There are two types of errors which are systematic errors and random errors.

3.1.1 Systematic Errors

The source of systematic errors is due to the imperfections of the measuring instruments. Systematic errors are fixed errors and for each measurement, this fixed error value is either subtracted or added.

For instance, when a voltmeter without its reset setting done shows 0.1 volts when not in use, then an excess value of 0.1 volts is taken into consideration in all measurements we conduct with the same voltmeter. If 0.1 volts are subtracted from all measurements, the correct measurement value is obtained. In carrying out an experiment, it is important to consider possible sources of systematic errors and to take precautions to eliminate them during the construction and calibration of the instruments.

3.1.2. Random Errors

These errors arise from unknown and unpredictable variations in the experimental situations. Random errors are generally very small and the chances of errors being positive or negative are equal. These random errors emerge due to the diversity of the mechanism and may not be completely removed (as mentioned in our example given for systematic errors) from the experiment results. These errors require statistical analysis because of their randomness. The best way to determine random errors is to repeat the same measurement more than once. However, we will use another method since it is not always possible to make repeated measurements: the error in a measurement device is equal to the smallest amount that device can measure.

Random Errors in Position Measurements

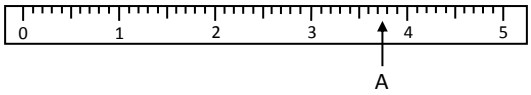


Figure-1: Measuring the position of A using a ruler.

Let us assume we are making a measurement with a ruler. The error in the ruler is equal to the smallest segment length on the ruler. For example, as seen in the **Figure-(1)**, we are measuring a position measurement of a point A with a ruler divided into "mm" segments and find the result to be 3.7 cm. A reading may appear in between points 3.7 and 3.8 cm.

Because the smallest quantitative value of our device can measure is 1 mm which is standard for all position measurements with this ruler, then, we write our result as $3.7 \pm 0.1\ cm$. Thus, we may mention that a ruler divided into millimeters is more reliable than one divided into centimeters.

Random Errors in Length Measurements

It is a measurement error in determining a distance between two points in one dimension.

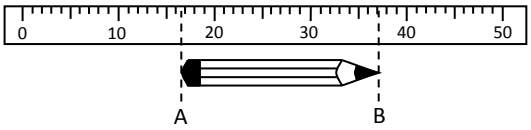


Figure-2: Measuring the length of a pencil using a ruler.

As an example, in the Figure-(2), let us measure the length of a pencil using a ruler. The length of the pencil is calculated by the difference between the end points of A and B. Therefore, the positions of these two points should be measured with respect to the same reference point. Each position measurements come with an error of 1 mm. Therefore, the error of this measurement is 2 mm. As seen in the figure, we can read the position of point A as 17 cm and position of point B as 37 cm with respect to the "0" point on the scale of the ruler. Then, we write our result as:

$$L \pm \Delta L = (37 - 17) \pm 0.2cm = 20 \pm 0.2\ cm.$$

As being experienced in this example, the effect of random errors can be reduced by improving the experimental technique and repeating the measurement many times and taking the average.

- ☑ The scale of a measuring instrument is not exact.
- ☑ The exact reading of a scale is impossible.
- ☑ The exact measurement of any quantity is impossible.
- ☑ No experiment gives the right answer.

3.2. Accuracy and Precision

Experimental errors are the difference between a measurement and the true value or between two measured values. Under the effect of these errors, the reliability of the measured values are analyzed by its *accuracy* and *precision*. Accuracy and precision are two important factors to consider when taking data from measurements. The accuracy of an experiment is a measure of how close the experimental result comes to the *true (or accepted) value*. It is related to a measure of the correctness of the result. Source of systematic errors determines how accurate the measurement is. The precision of an experiment is a measure of how reproducible the result is. That is, it is a measure of the magnitude of the uncertainty of the result due to random errors. It is related to a measure of repeatability of the result and therefore, it is usually limited by the sources of the random errors.

Both accuracy and precision reflect how close a measurement is to an actual value, but accuracy reflects how close a measurement is to a known or accepted value, while precision reflects how reproducible measurements are, even if they are far from the accepted value.

To clarify these definitions, consider an example about throwing and striking a bullet to a target (Figure-3).

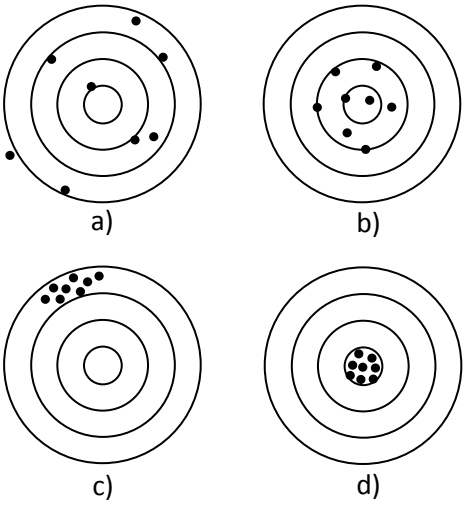


Figure-3: The difference between accuracy and precision.

In the **Figure-(3),** in the Part-(a), there is an example showing an example of striking which is neither precise nor accurate. Because the aim of throwing the bullets is to hit the target on its center, they are not in the middle of the inner circle and also the differences between them are very large. Part-(b) shows an example of striking which is accurate but not precise. As compared to the results in part-a, the bullets are near the centre of the target but they are not close to each other. Part-(c) shows an example of striking which is precise but not accurate. As seen in Part-(c), the bullets are in a group which means the results are approximately the same; the difference between the bullets is very low. However, they are not located near the centre of the target. Lastly, Part-(d) shows an example of striking which is both accurate and precise. This means that each of the trial gives the true (accepted) result, the bullets are located near the centre of the target, and also they give the similar results, the difference between the bullets is small.

The accuracy of an experiment often depends on both systematic and random errors. The precision of an experiment typically depends on random errors.

4. Significant Figures

When certain quantities are measured, the measured values are known only to within the limits of the experimental uncertainty. The number of significant figures in a measurement can be used to express something about the uncertainty. The number of significant figures is related to the number of numerical digits used to express the measurement.

The degree of certainty of a measurement is implied by the way the result is written or reported. While reading the value of an experimental measurement from a calibrated scale, only a certain number of digits can be obtained or read. Significant figures are those figures which are known to be reasonably trustworthy; they are figures actually obtained from the measuring instruments, including zeros and estimated figures.

The position of the decimal point does not affect the number of significant figures. If a zero is used to locate the decimal point, it is not significant, but if it actually represents a value that is read on the instrument, or estimated, then it is significant.

Give the figures you read from a scale with only the last figure estimated as a result of measurement. Never put extra figures. The digits including the last one estimated are called significant figures.

The number of significant figures in a measurement is determined as follows:

- ☑ The leftmost nonzero digit is the most significant.
- ☑ If there is no decimal point, the rightmost nonzero digit is the least significant.
- ☑ If there is a decimal point, the rightmost digit is the least significant, even if it is zero.
- ☑ All digits between the least and most significant digits are considered to be significant.

Pay attention to zeroes whether they are significant or not.

As an example, the numbers in the following list of numbers all have four significant figures. There is an explanation for each.

- ✓ 6739: All four nonzero digits are significant.
- ✓ 456900: The last two zeros are not significant because there is no decimal point.
- ✓ 0.008589: Zeros at the left are never significant.
- ✓ 0.01000: The zero at the left is not significant, but since there is a decimal point, the three zeros at the right are significant.
- ✓ 10.60: There is a decimal point, so all four numbers are significant.
- ✓ 1.057: Again, there is a decimal point, so all four are significant.

4.1. Scientific Notation

Physics deals with some very large and very small numbers. The scientific notation is a convenient method to express numbers rather than conventional notation. Scientific notation also allows numbers to be expressed in a form that clarifies the number of significant figures. The exponent in scientific notation is equal to the number of times that the decimal point must be moved to produce a number between 1 and 10.

Write the result of a measurement as two factors; the first factor contains all the significant figures, having one nonzero digit in front of the decimal point, and the second factor is a power of 10.

Examples for the usage of this notation are given below;

- ✓ $255\,mm = 2.55 \times 10^2\,mm$
- ✓ $11.1\,cm = 1.11 \times 10^1\,cm$
- ✓ $0.0125\,km = 1.25 \times 10^{-2}\,km$
- ✓ $360\,g = 3.60 \times 10^2\,g$

4.2. Calculation with Significat Figures

The calculations carried out with significant figures usually produce extra figures as a result of calculation. It is important that the calculations should be consistent with the values that have measured. The value with the least significant digits sets the standard for the degree of accuracy for the measurements. This means that the result calculation should have only one estimated figure.

4.2.1. Rounding Numbers

The result of the measurements can be rounded off to the desired number of significant figures before they are used in calculations. The rules for rounding of the numbers are:

- ☑ If the last figure is less than 5 (0, 1, 2, 3 or 4), it should be dropped and the others should be unchanged.
- ☑ If the last figure is equal to 5 or greater (5, 6, 7, 8 or 9), it should be dropped and the numerical result should be increased by 1.

For instance,

- ✓ 2.431 may be rounded off to 2.43 when rounded off to the nearest hundredths place.
- ✓ 2.431 may be rounded off to 2.4 when rounded off to the nearest tenths place.
- ✓ 2.431 may be rounded off to 2 when rounded off to the nearest units place.
- ✓ 6.587 may be rounded off to 6.59 when rounded off to the nearest hundredths place.
- ✓ 6.587 may be rounded off to 6.6 when rounded off to the nearest tenths place.
- ✓ 6.587 may be rounded off to 7 when rounded off to the nearest units place.

4.2.2. Multiplication and Division

In the multiplication or division of the numerical results, the result should be rounded off to the least number of significant figures used in the calculation. For example,

```
   3.519                          3.5 (rounded off)
 x 2.3 (least significant)      x 2.3
 -------                        -----
   10557                          105
+ 7038                         +  70
 -------                        -----
 8.0937 (≈8.1)                  8.05 (≈8.1)
```

As can be seen from the example, since the least number of significant figures is 2, the result is rounded off 2 significant figures.

The same result can be obtained by rounding off the all multipliers to the least number of significant figures before the multiplication.

$$\frac{102.35}{15.72} \Rightarrow \frac{10235}{-9432}\,\Big|\,\frac{1572}{6.5108\ (\approx 6.511)}$$

$$8030$$
$$-7860$$
$$1700$$
$$-1572$$
$$12800$$
$$-12576$$
$$224$$

The example above shows a division of two measured values. Similar to the multiplication case, the number of significant figures of the result should be equal to the number of significant figures of the value with the least number of significant figures. Therefore, the result has 4 significant numbers.

In the multiplication or division of numerical measurements, retain in the result only as many figures as the number of significant figures in the least accurate number used in the calculation.

4.2.3. Addition and Substraction

When the numerical results are added or subtracted, the answer can contain no more decimal places than the least accurate measurement. This means that the number of places after decimal point in the answer should be less than or equal to the number of decimal places in each term in the calculation. For example,

$$\begin{array}{r} 0.073 \\ 13.7 \\ +\ 156 \\ \hline 169.773\ (\approx 170) \end{array}$$

The same result can be obtained by rounding off the all terms to the least accurate terms in the calculation before addition.

When carrying out addition or subtraction with numerical measurements, do not carry the result beyond the first column (from the left) that contains a doubtful digit.

5. Errors in Calculated Results

In the real case measurements, obtained physical quantity should be written with errors of the measurement. Whenever the measurements are used in calculations, the error should be used in these calculations to move each uncertainty of the measurements to the final result.

To work on the usage of errors in the calculations, assume that there are two recorded measurements as $x \pm \Delta x$ and $y \pm \Delta y$.

5.1. Addition and Substraction

In these calculations, the maximum possible error ΔR in the final result R is the addition of errors from these measurements.

For addition:

$$R \pm \Delta R = (x \pm \Delta x) + (y \pm \Delta y) = (x + y) \pm (\Delta x + \Delta y)$$

For subtraction:

$$R \pm \Delta R = (x \pm \Delta x) - (y \pm \Delta y) = (x - y) \pm (\Delta x + \Delta y)$$

When measured quantities are added or subtracted, the maximum possible error in the result is the sum of the errors in the quantities used in the calculation.

5.2. Multiplication and Division

The multiplication and division of two measured data is a little complicated than the previous calculations. They are performed as follows:

For multiplication:

$$R \pm \Delta R = (x \pm \Delta x).(y \pm \Delta y) = x.y \pm x.y\left(\frac{\Delta x}{x} + \frac{\Delta y}{y}\right)$$

In this equation, the first term is the product of the actual measurements and second term is its corresponding error.

For division:

$$R \pm \Delta R = \frac{(x \pm \Delta x)}{(y \pm \Delta y)} = \frac{x}{y} \pm \frac{x}{y}\left(\frac{\Delta x}{x} + \frac{\Delta y}{y}\right)$$

In this case, the first term is the division of the actual measurements and second term is its corresponding error.

6. Expressing Errors (Statistical Overview)

6.1. Average (Mean) Value

As mentioned in the types of errors part, random errors can be minimized by repeating the measurements several times. It is expected to obtain similar results for all trials. For this set of measurements, the actual result is most probably obtained by the average value. The average or mean value of a set of n measurements can be expressed as:

$$\bar{x} = \frac{x_1 + x_2 + x_3 + \cdots + x_n}{n}$$

6.2. Average Deviation of the Mean

Taking the average value of a series of measurements due to the random errors, there is a difference between each result of the measurements from the average value. This deviation d_i of any single measurement x_i is written as:

$$d_i = x_i - \bar{x}$$

When using the average value of the results, the best estimated error is defined as the average deviation of the mean (A.D.):

$$A.D. = \frac{d_1 + d_2 + d_3 + \cdots + d_n}{n}$$

6.3. Standard Deviation

The precision of the measurements can be determined by calculating a quantity called standard deviation from the mean of the measurements. The standard deviation represents the average of the square of the variation of individual measurements about the average value. It is defined as:

$$\sigma = \sqrt{\frac{d_1^2 + d_2^2 + d_3^2 + \cdots + d_n^2}{n}}$$

6.4. Absolute, Fractional and Percentage Errors

The error in a measured quantity may be expressed either as the absolute error, the fractional (or relative) error or percentage error. The absolute error is defined as the magnitude of difference between actual and measured value and it has the same units as the measured quantity. For example, a distance measurement made with a ruler marked in millimeters can be expressed as $25.5 \pm 0.2\,cm$. The absolute error in this quantity is $\pm\, 0.2\,cm$.

Fractional error is the ratio of the absolute error to the quantity and it has no unit. It can be expressed as:

$$Fractional\ error = \frac{Absolute\ Error}{Actual\ Value}$$

It is commonly expressed as a percentage error of a measurement to give the error to the actual value multiplied by 100.

$$Percentage\ Error = Fractional\ Error \times 100\%$$

The number of decimal places of the error part should be same with those of measurement.

For example,

- 25 ± 1 (true)
- 13.25 ± 0.05 (true)
- 25.2 ± 0.12 (false)
- 12.5 ± 0.5 (true)

7. Graphs

Graphing is a way of displaying relations between various quantities, parameters, or measurable variables in the measurements for clarity. Basically, it is a visual picture of the quantitative relationship between the physical quantities defining how one quantity changes if another quantity that is related to it also changes in the measurements.

Generally, in basic physics experiments, two-dimensional graph is enough to represent the relation between two related variable physical quantities. In these types of graphs, the horizontal axis (*x-axis* or *abscissa*) is used for the *independent variable*, and the vertical axis (*y-axis* or *ordinate*) is used for the *dependent variable*. For example, in a free fall experiment, assume that the measurement is done to determine the height *h* from the ground of a ball falling freely. In this case, the dependent variable *h* is investigated with the independent variable time *t*.

Moreover, in all plotting procedures, one of the most important points is to define the location of the data points obtained from the measurement according to the reference point called as the origin. It means that, any data point in the two-dimensional graph can be symbolized as (*x, y*) which refers to its coordinates on the *x* and *y* axis, with respect to the origin (0, 0).

Because the aim of usage of the graphs is summarizing a lot of information into one picture, a graph should be self-explanatory. The guide lines of graphs expecting from the experimental works are:

- Write the **title** of the graph
- Show x and y **labels** clearly
- Specify **units** on the axis
- **Scale** the axis correctly
- Identify the **data points** clearly
- Show the **error bars**
- Draw **a best** and **a worst line**
- Show equation of the **slope** and **y-intercept**

Title of the graph

- Each graph should have a title. The title is related to the variables that will be plotted. It is very important that someone should figure out easily what is plotted. A title is usually written for example “Distance (**D**) versus Time (**t**)” or “Voltage (**V**) versus Current (**I**)”.

Label and the Unit of the Axes

- ***x*** and ***y***-axis should be labeled according to the experimental data. Labels on the axes should have a unit. For instance, for the ***x***-axis which is used for the independent variable can labeled as “Time (seconds)” and ***y***-axis can “Distance (meter)”.

Scaling of the axis

First of all, the data numbers should be observed in order to determine how big a graph to make. Scaling should be appropriate so that the data points are distributed as widely as possible over the area of the graph paper. The independent variable lies on the *x*-axis and dependent variable lies on the *y*-axis. The data values are examined separately to fit the axes. The scalings may be different on each axis and does no necessarily need to start from zero. Each axis must be divided into equal parts in itself. The starting and the end points of the axis are determined by subtracting the error part from the minimum actual data and adding the error part to the maximum actual data. This method is done for each variable and start points from the origin can vary.

Identifying the Data Points

Each row of the data table indicates a data point. If there are five data values for each dependent and independent variable, then five data points should be identified on the graph.

Error Bars

Error bars represent the uncertainity of each data points. While drawing the error bars on the data points, the scaling on each axis is considered and the calculations for the error bars should be similar to the scaling calculations.

Up to these steps, the graph should be similar to the one given below in the **Figure-(4)**.

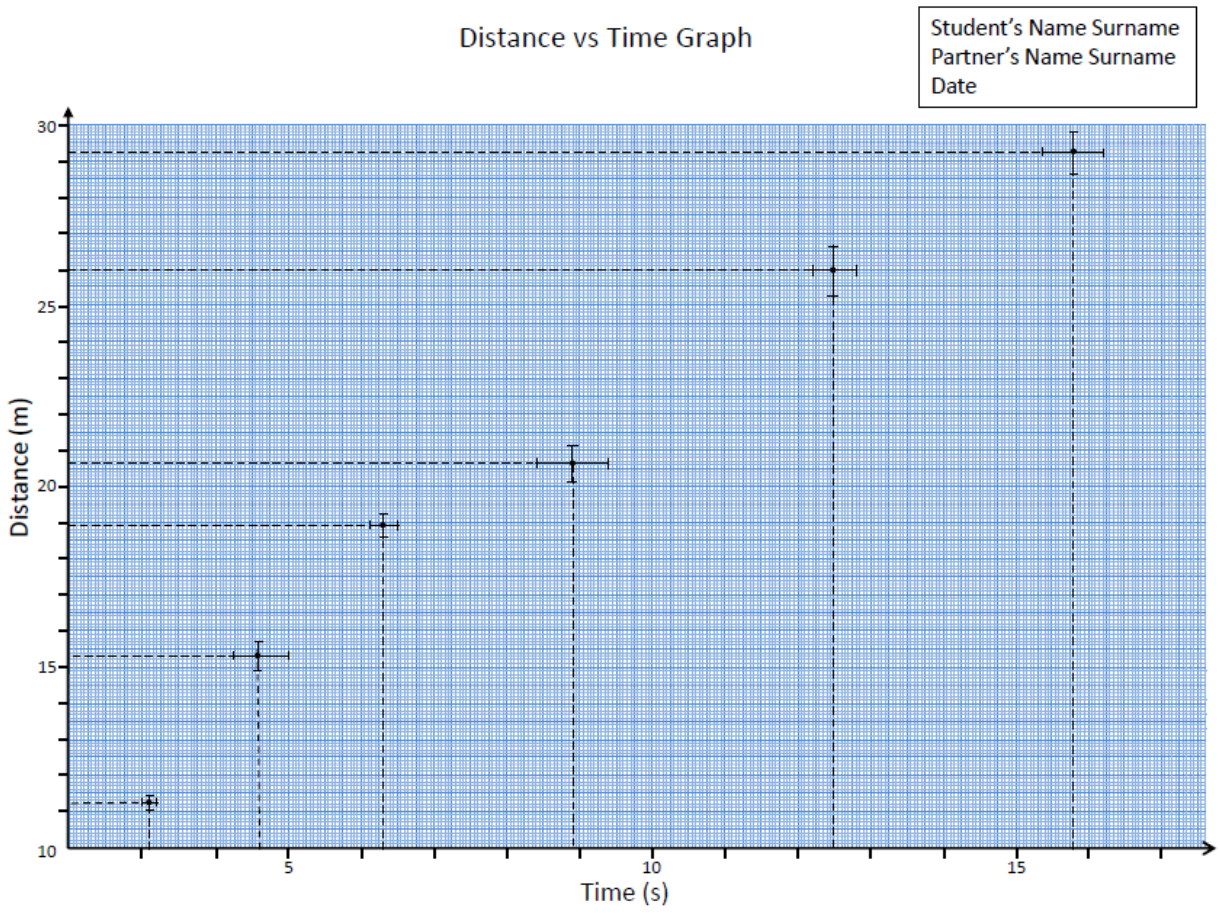


Figure-4: A graph of Distance vs Time.

7.1. Analysis of a Straight Line Graph

Two linearly related quantities, x and y, have an algebraic equation of the form:

$$y = mx + b$$

where m and b are constants. $\boldsymbol{y}$ vs $\boldsymbol{x}$ graph gives a straight line with slope m, which usually corresponds to a physical quantity in the measurements.

Assume that there are two points (x_1, y_1) and (x_2, y_2) in a data set of a measurement that shows a straight line behavior when plotted using $\boldsymbol{x}$ and $\boldsymbol{y}$ axis (Figure-4).

The changes in x and y axis between these points are defined as $\Delta x = x_2 - x_1$ and $\Delta y = y_2 - y_1$, respectively.

Slope, $\boldsymbol{m}$ is expressed as;

$$m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

In slope calculations, only one data point on the line cannot be chosen. Determining any arbitrary two points on the line is enough to calculate the slope. Because measured points contain an error part, we are required to draw a best straight line fitting to the data points to calculate the best value of the slope m. A best line is a straight line that best represents the data on the graph.

This line may pass through all, some or none of the data points. The main important point is that an equal number of points lie on each side of the best line.

The best straight line, however, must pass inside as many error bars as possible.

In order to calculate the error on ***m***, we need a line that worst fits to the data points. It must be chosen to give the maximum possible deviation from the best line. The drawn line must pass through the **error bars** of the first and last data points and the worst line must cross the best line.

Best and worst lines cannot be parallel.

The ***b*** in the equation is called the **intercept** and it gives the point at which the straight line crosses the *y*-axis. Simply, when $x = 0$, then $y = b$.

Actually in the analysis of straight line graphs, the aim is to determine from the graph the slope as $m \pm \Delta m$ and the intercept as $b \pm \Delta b$, where Δm and Δb are the maximum possible errors in slope *m* and in intercept *b*, respectively.

These errors can be easily determined by drawing the *worst possible straight line* on the graph. The analysis of this worst possible line is also done in the same way to obtain the worst possible slope *m'* and the worst possible intercept *b',* and then we get the possible errors Δm and Δb as:

$$\Delta m = |m - m'|$$

and

$$\Delta b = |b - b'|$$

After drawing the best and worst lines, the calculation steps of slope of best and worst line should be shown on the graph as in the example given below in the **Figure-(5)**.

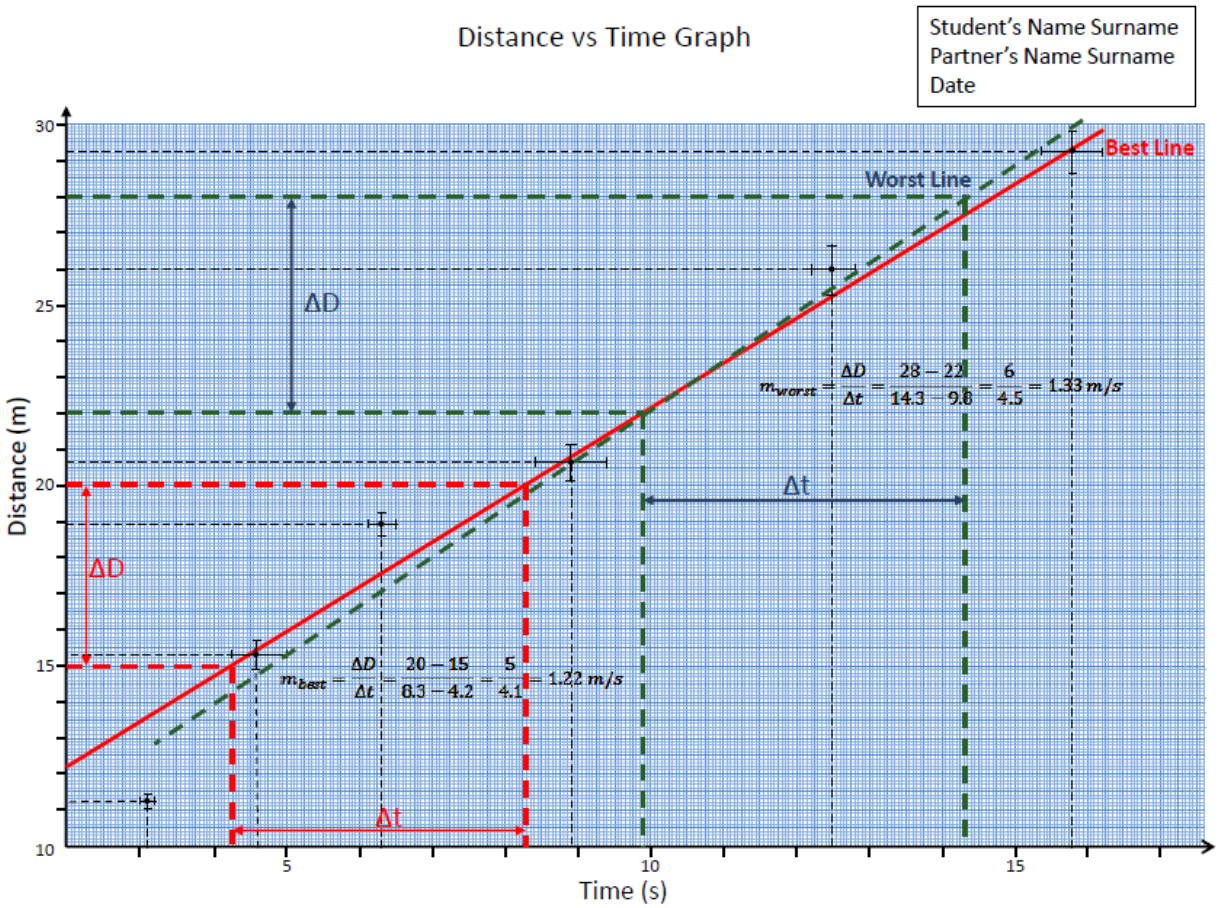


Figure-5: Slope calculations on the graph.

7.2. Nonlinear Graphs

In nature, the simplest relation to carry out in the experiments is direct relation as stated in the straight line analysis. However, the other relations are often valid in our daily life.

7.2.1. Inverse Proportion

This graph states that x is inversely proportional to y. This inverse relation can be rewritten as:

$$yx = k$$

or,

$$y = k/x$$

where $\boldsymbol{k}$ is a constant.

In such a relation, when $\boldsymbol{y}$ versus $\boldsymbol{x}$ graph is plotted, it gives a hyperbolic relation as shown in the **Figure-(6)**.

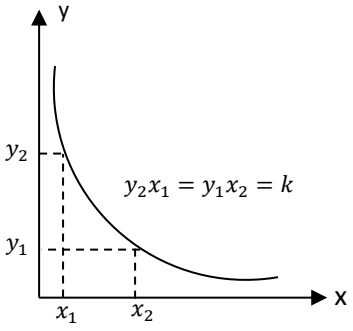


Figure-6: $\boldsymbol{y}$ versus $\boldsymbol{x}$ graph.

However, if the x-coordinate, independent variable is chosen as $1/x$, and then, if $1/x$ versus y graph is plotted (Figure-7), the direct proportion can be obtained.

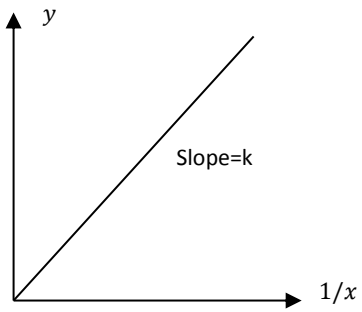


Figure-7: y versus $1/x$ graph.

Therefore, the experimental data can be studied as to be straight line graph.

7.2.2. A Square Law Graph

A square law relationship between any two variable *x* and *y* can be given as

$$y = kx^2$$

where *k* again a constant. In this relation, ***y*** is related non-linearly to the ***x***. Note that, instead of working on ***y*** versus ***x*** graph (Figure-8), it is also convenient to use a graph of *y* versus x^2. This graphical representation gives a straight line whose slope would be ***k*** as shown in the Figure-(9) and it would pass through the origin.

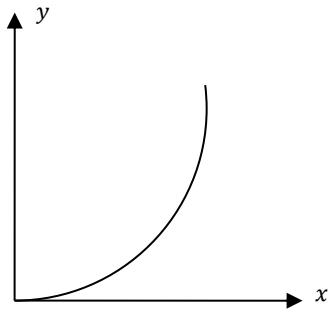


Figure-8: y versus x graph.

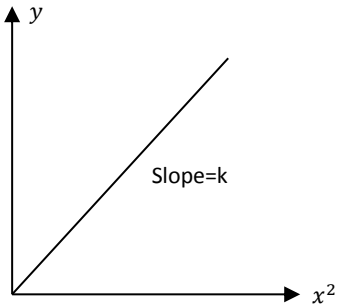


Figure-9: y versus x^2 graph.

7.2.3. An Inverse Square Law Graph

An inverse square law can be described as

$$y = k/x^2$$

where ***k*** again a constant. In this relation, the variable *y* is depending nonlinearly on ***x***, but it is depending linearly on $1/x^2$. Therefore, if the *x*-axis is chosen as $1/x^2$, the corresponding plot gives a straight line with a slope of ***k***.

7.2.4. Exponential Relation

If a graph of ***x*** and *y* has an equation given by

$$y = ke^{cx}$$

where ***c*** and ***k*** are some constants and *e* is the base of natural logarithm and equal to 2.718, taking the logarithm of both sides of this equation gives a straight line relation as

$$lny = lnk + cx$$

When lny is plotted as a function of x, the obtained straight line graph gives the slope as $\boldsymbol{c}$ and the intercept lnk (see **Figure-10**).

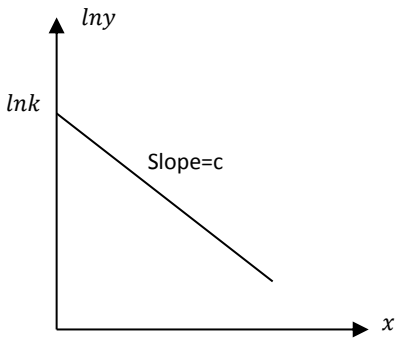


Figure-10: lny versus x graph.

Instead of using natural logarithms to the base 2.718, logarithms to the base 10 can also be used to get a straight line graph. The corresponding equation can be written as

$$logy = logk + cx$$

and the graph $logy$ versus x gives the slope as $\boldsymbol{c}$, and the intercept as $logk$.

7.2.5. Power Laws

Two variables $\boldsymbol{x}$ and $\boldsymbol{y}$ are related by the general equation,

$$y = kx^c$$

where $\boldsymbol{k}$ and $\boldsymbol{c}$ are constants. When taking the logarithm of both sides, the equation becomes

$$logy = logk + clogx$$

where it is in a straight line characteristics.

When $logy$ is plotted as a function of $logx$, obtained straight line gives the slope *c* and the intercept $logk$.

Instead of using logarithms to the base 10, natural logarithms to the base 2.718 can also be used to get a straight line graph. The corresponding equation can be written as

$$lny = lnk + clnx$$

and the graph lny versus lnx gives the slope as $\boldsymbol{c}$, and the intercept as lnk.

8. Experimental Set-up



Figure-11: Aluminum protective case for the measuring instruments.



Figure-12: The experimental set-up.

8.1. Ruler

A **ruler** is one of the most common measuring instruments to measure distances and to draw straight lines. Depending on the experimental requirements, there are many rulers with different sizes and shapes.

When measuring the length of an object, the measurement should not be taken from zero on the ruler, non-zero origin can also be chosen as a starting point. Reading a value from a ruler involves lining up the two points (one of the end point on the object and the reading point on the ruler) with the marks on the ruler, however, exact lining up is impossible. The apparent distance between two points on the object, and therefore the values of the reading, depends also on the position of your eye. Note that there are divisions on the ruler, and the *smallest* division is one millimeter. So, you should make a best *estimation* on a reading between the millimeter markings on the ruler. The position of a point near a ruler may appear different if viewed from the left or right of a line of sight perpendicular to the ruler, named as *parallax*. For example, thediagram shows the position of an object near the ruler.

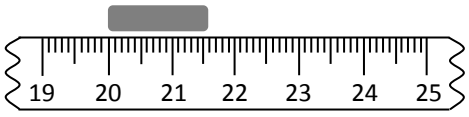


Figure-13: Measuring the length of an object.

Using the position of the ruler as shown in the **Figure-(13)**, the two edge points of the object should be estimated. While doing the measurements, be aware of the thickness of the millimeter lines on the ruler and the parallax effect on your observation. Therefore, for a one millimeter smallest division on the ruler, the estimation limits of error is equal to itself (1 mm). Then, the positions of the two end points of the object on the ruler can be written as 20.0 and 21.6 cm. Note that, the result of the measurements must be written with the correct number of significant figures and the unit of measurement.

As a difference between these two ends, the length of the object can be recorded as:

$$d = 1.6 \pm 0.1\ cm \text{ or } d = 16 \pm 1\ mm$$

The significance of $\pm\ 1\ mm$ (or $\pm\ 0.1\ cm$) is that when we repeat our measurement several more times, the readings we obtain are expected to be one of the values 15.0 mm, 16.0 mm or 17.0 mm. For many repetitions of the measurement, the value most frequently occurring will be the value 16.0 mm.

8.2. Caliper

A (vernier) caliper is a common tool used in laboratories and industries to accurately determine the fraction part of the least count division. Unlike a ruler, the vernier caliper allows the fractional part of the smallest division to be accurately determined, not merely estimated. It is a standard method used in instruments to get higher precision readings. It can be used to measure length of an object, the outside/inside diameter of a hole and the depth of a hole.

The schematic diagram is illustrated in **Figure-(14)**. As seen in this figure, the main parts of a vernier caliper are a main scale, a ***Vernier*** scale and a fixed and sliding jaw.

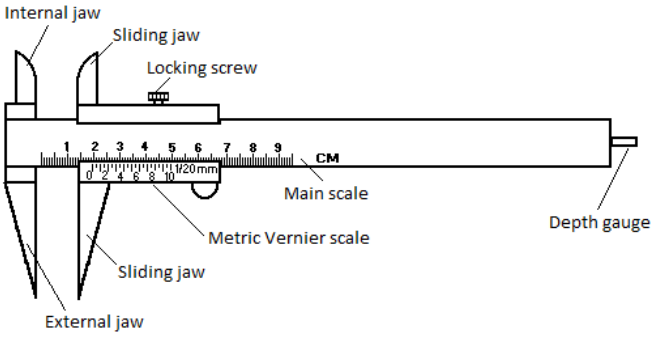


Figure-14: ***Vernier*** caliper.

Vernier calipers are available with metric and imperial graduations. Also, there are different types of caliper may employ other types of readout such as, a dial for least count readout or an electronic digital readout.

Calipers are available in various sizes, with measuring ranges from 100 mm to 3000 mm. Generally, calipers that have a measuring range of 300 mm or less are classified as small calipers and ones with larger ranges are classified as large calipers. Typical measurement resolution for a vernier caliper is 0.05 mm. This uncertainty in a measurement is changed depending on its type; it is 0.02 mm for dial calipers and 0.01 mm for digital calipers.

Using a caliper

The vernier consists of a main scale engraved on a fixed ruler and an auxiliary scale engraved on a moveable jaw (see **Figure 14**). The moveable jaw is free to slide along the length of the fixed ruler. The main scale is calibrated in centimeters with the smallest division in millimeters. The moveable auxiliary scale has 10 divisions that cover the same distance as 9 divisions on the main scale. Therefore, the length of the auxiliary scale is 9 mm. When the vernier is closed and properly zeroed, the first mark (zero) on the main scale is aligned with the first mark on the auxiliary scale. The last mark on the auxiliary scale will then coincide with the 9-mm mark on the main scale.

External Measurement

These types of measurements are the most basic usage of the calipers. In this measurement, at first, the jaws should be opened enough to place the caliper over the object to be measured, and then the movable jaw should be slid toward the fixed one until it contacts the object. Therefore, to measure length, place the object between the jaws of the caliper and lightly close the jaws on the object.

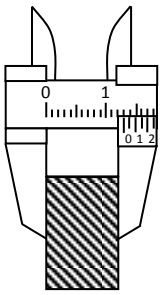


Figure-15: Measuring the length of the object.

However, there may be some errors that caused the result not to be accurate. The most effective source of error is occurred when the object is in contact with the jaws such as the characteristics of the objects (whether it is soft or hard), or your pressure to put the object between the jaws. Moreover, if the calipers are not straight (that is, the jaws are not in contact with the contact sides of object) or if there is dirt on the jaws or the contact points, the measurement may not be accurate.

Internal Measurement

To determine the internal diameter of a piece of object as shown in the **Figure-(16)**, the smaller jaws on the top of the caliper are used. In this measurement, at first, the jaws should be closed and placed into place to be measured. Then, taking the internal jaw is fixed at one end, movable jaw should be moved until it contacts with the other end point of the object. The locking screw should be tightened so that an accurate measurement can be made.

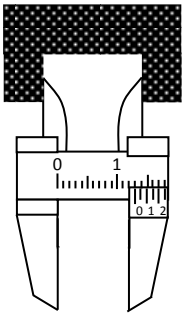


Figure-16: Measuring the inner diameter of the object.

In such type measurements, the most common problem is keeping objects lined up correctly when you are taking an inside measurement.

Depth Measurement

For the ***depth*** measurements, the depth gauge of the caliper is used. When sliding the movable jaw, this part extends out of the far end. In this measurement, initially one end of the caliper should be placed against top of the hole as shown in the **Figure-(17)**.

Then, using movable jaw, the caliper should be opened by sliding the movable jaw towards the hole until the depth gauge contacts the bottom of the hole. The locking screw is tightened and the measurement can be read on the scales of the caliper.

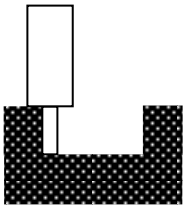


Figure-17: Measuring the **depth** of the object.

Measuring the **depth** of a hole by a caliper is difficult than the other types of measurements. Since caliper's depth gauge must be forced into contact with the bottom of the hole, in some cases it is difficult to determine the contact point and also to contact with the bottom point accurately.

Reading Calipers

On the caliper, there are two scales, a main (or fixed) scale and a Vernier (or sliding) scale. In typical metric calipers, the numbers on the main scale are given in centimeters and therefore the smallest division on that scale is a millimeter. Moreover, the Vernier scale is divided to 10 points from 0 to 10. Note that, when reading the Vernier scale, you start from the 0 on the sliding scale.

After the measurement is done, at first, the position of 0 point of the Vernier scale on the main scale is read. It is a rough estimation of the length of the measuring part of the object. Then, this **0** point should be observed whether it is in the same line with the mark on the main scale or not. If it is in the same positioning, the reading from the main scale can be taken as the result of the measurement with the instrumental uncertainty. On the other hand, if not, the line on the Vernier scale aligns with a line on the main scale must be found to get a correct reading result. This observed point must be evaluated considering the Vernier scale contains a hundred points.

In this case, the point is read on this scale in tens digit. This gives a reading for the hundredths place from the decimal point of the main scale reading. Again, the result of the measurement should be written with expressing the instrumental uncertainty. The following examples on an integer and non-integer readings explains how a reading is taken using a metric vernier caliper.

The **Figure-(18)** below shows the case where the zero graduation on the vernier scale is aligned perfectly with a graduation on the ***main scale***. In this case, the zero on the Vernier scale and the 35 mm graduation on the main scale is aligned. The correct caliper reading is therefore 35.00 mm.

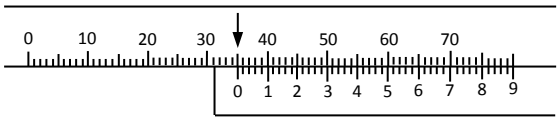


Figure-18: An integer reading on the main scale.

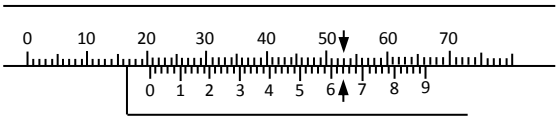


Figure-19: A non-integer reading on the main scale.

Figure-(19) illustrates the case where the ***zero*** on the Vernier scale does not line up with a graduation on the main scale. In this case the procedure for finding the correct reading is as follows.

1. Look along the main scale and determine the smaller of the two graduation numbers that most closely line up with the zero on the vernier scale. In the example above figure, the zero on the Vernier scale lies between 20 and 21 on the main scale. The correct main scale reading to use is therefore 20 mm.

2. Now look along the vernier or sliding scale and see which graduation lines up with a main scale division. In figure, this is 54 on the vernier scale, and note that the resolution of this scale is 0.01 mm.

3. The correct reading in the above example is therefore 20 mm + 54 x 0.01 mm = 20.54 mm.

Note that there is no ambiguity in the reading of the caliper so the reading uncertainty is zero. However, there is an instrumental uncertainty of ± 0.01 mm.

Taking several measurements on any one dimension and recording the mean can give extra confidence in measurements and is necessary for estimating the uncertainty of measurement.

Sources of the Errors

There are a number of factors affecting the measuring accuracy of calipers. The user should be aware of both the cause and effect of these errors. As an error source to the construction of the caliper, straightness of the caliper jaws are important to control before usage. These inside and outside measuring faces should be perfectly flat and also be parallel to one another.

Moreover, resolution of the Vernier scale is the other source of error that affects the accurate reading from caliper. As in the ruler usage, because in most of the measurements, the main scale and Vernier scale of a caliper are not in the same plane, the apparent alignment position can vary depending on the parallax error. Most of the calipers are not provided to a constant measuring force to the measuring object during the measurement. Therefore, the measuring force will vary each time when a measurement is made, especially with different users.

In addition to these, before taking a measurement, the surfaces of jaws and depth gauge, and also the measured faces of the object should be cleaned to eliminate measurement errors.

8.3. Micrometer

A micrometer, similar to a caliper, is a measuring instrument for precise measurement of a very small objects and/or distances. This device is based on a graduated screw mechanism to produce precise linear displacement of a spindle along its axis.

Distance measurement is achieved referencing the linear displacement of the spindle to a fixed measuring face on the axis of the spindle (the anvil). Figure-(20) shows the main components of a micrometer.

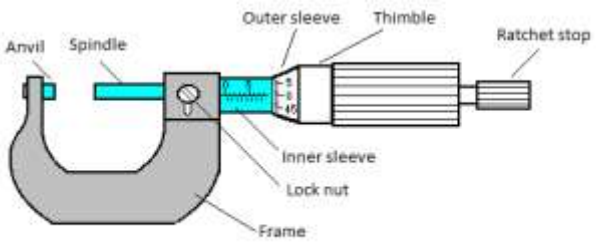


Figure-20: Components of the micrometer.

Like calipers, micrometers are generally used to measure the outside and inside diameters, and also the depth of the holes in an accurate way. In a micrometer usage, spindle and thimble reading lines are the parts enable to get reliable results in detail. A standard micrometer has a screw thread of 0.5 mm pitch with a thimble graduated in fifty equal divisions around its circumference. There are many micrometers that produced in different size ranges. In a typical way, as a smallest one 0 – 25 mm size range is available, and the others are formed in a 25 mm increment in size ranges. The choice is dependent on user's needs.

In order to get a precise measurement, the most effective case in micrometer usage is the force that spindle applies to the measuring object. The applied force varies the measurement result due to the deformation on the spindle or spring in a micrometer even if it does not affect the surface of the object. Therefore, a constant and same force should be applied to the object to minimize the variation in measurements and also prevent instrumental errors. In a micrometer, the ratchet stop is commonly used to facilitate this setting. The mechanism comprises two ratchets that are opposed to each other and whose toothed faces are pressed together by the force of a spring. When the barrel of the ratchet stop is rotated clockwise, both ratchets turn together until the measuring force reaches to a certain limit.

As the force exceeds this limit, the ratchet in the barrel spins idly and clicks. When the barrel is rotated counter-clockwise there is no idling between the ratchets because the ratchet teeth remain engaged and the thimble is allowed to rotate.

Reading a micrometer

To perform a measurement with a micrometer the initial procedure is that the anvil should contact with the surface of the measuring object. Then, with rotating the thimble or the ratchet until the spindle touches the object, the anvil and spindle should be close to (but not in contact with) the object.

The ratchet should be rotated until you hear a click from the ratchet stop to apply a pressure to keep the object in between anvil and spindle.

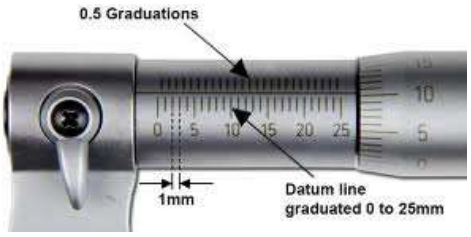


Figure-21: Reading a value from the sleeve scale of a micrometer.

In reading a measurement result on a micrometer scale, at first, read the number on the major divisions of the sleeve scale. Then, a further half mm measurement can be seen on the bottom half of this scale. After these two readings, the thimble scale must be read to complete the reading procedure. In this final step, the value of the line on thimble scale that coincides with the datum line on the sleeve is added to the sleeve scale reading. With 0.01 mm discrimination micrometers, there is an instrumental uncertainty of $\pm\, 0.01\ mm$.

An example of ***reading*** a micrometer is shown in the Figure-(22).

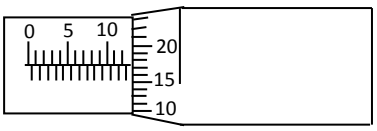


Figure-22: Micrometer reading.

1. The major sleeve scale reading is 12 mm.

2. A further half mm measurement seen on the bottom half of the scale gives an addition of 0.5 mm. Therefore, measurement reading on the sleeve scale is 12.5mm.

3. The line on the thimble scale coincides with the datum line shows 16 full divisions (these are hundredths of a mm). Therefore, the correct reading in the above example is 12.5 mm + 16 x 0.01 mm = 12.66 mm.

4. Finally, the uncertainty in the micrometer reading should be added to the result as 12.66 ± 0.01 mm.

Taking several measurements on any one dimension and recording the mean can give extra confidence in measurements and is necessary for estimating the uncertainty of measurement.

Sources of Errors

At first, it is important for the user to be aware of the possible error sources in the measuring instrument, micrometer. The calibration and mechanical facilities should be checked before usage. Moreover, the surface characteristics of the measuring object affects the final result. It is because of the variations in contact points of object surfaces with anvil and spindle. The accuracy in result is obtained by the thimble scale reading. Since the reading is somehow estimation, the minimum graduation on this scale determines the uncertainty of the result.

8.4. Graduated Cylinder

Graduated cylinders are used to measure volumes accurately. It can be used to calculate the density of an object if its mass is known. Graduated cylinders usually are made from plastic or borosilicate glass. Common sizes are 10, 25, 50, 100, 250, 500, 1000 ml.

Water Immersion Method for Volume Determination

The volume of a solid object can be measured by using the water displacement method. In this method, the object is placed in a liquid that is less dense than the object. The addition of the object to the liquid raises the level of the liquid because the two volumes are additive.

This experiment assumes that the solid doesn't react with or dissolve in the liquid. V_1 is the initial volume of the liquid. V_2 is the sum of the volume of the liquid and the solid object. Notice that V_2 is greater than V_1. This is because the volume of the liquid and the volume of the solid object are additive. The volume of the solid object is (see **Figure-23**):

$$V_{solid} = \Delta V = V_2 - V_1$$

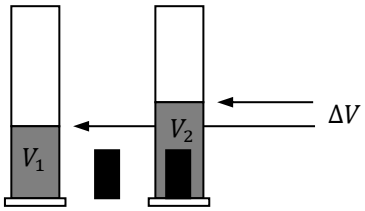


Figure-23: Volume determination by a graduated cylinder.

Measurements can sometimes be difficult to determine. The following are some important techniques. While measuring level of the liquids that have a curve at the surface, volume should be measured from the bottom of the meniscus. The meniscus is the curve formed at the surface of a liquid due to attraction of the liquid for the sides of the container (adhesion).

Measuring from the bottom in **Figure-(24)**, you should get $54 \pm 1\,ml$.

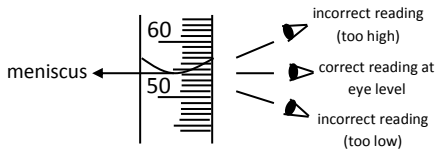


Figure-24: Reading a meniscus. Read the bottom of the minuscus while holding at eye level.

In our case, the size of the graduated cylinder is maximum 250 ml. The scale of the cylinder varies between 50 and 250 ml with an increment of 10 ml. If the volume of a solid is under 50 ml, you should start with at least 50 ml as an initial liquid volume.

Note that, fill the graduated cylinder with a liquid volume in which the measured object is totally sinked. Moreover, be careful about the maximum limit of cylinder for the total (liquid+object) volume.

In general case, uncertainty of the graduated cylinder is taken as 1 division on the scale. However, in our graduated cylinder, the smallest scale is 10 ml and it is a very huge value to use as an error of the measurement. Therefore, it is recommended that a half of the division can be used as an uncertainty of this instrument.

9. Experimental Procedure

In the first part of the experiment, we will try to calculate the densities of the objects with different shapes by measuring their volumes and masses. The volume measurements will be carried out by using four different measuring instruments; ruler, caliper, micrometer and graduated cylinder. Two different device; equal arm balance and digital balance, will be also used to determine the mass of these objects. To get a precise result, three measurements will be done to determine the average dimensions (in order to calculate their volume) and massof each object. The number of trials should be determined by the number of group members. Each group member should take one measurement because making another measurement will be under the effect of the previous measurement. Then, the results of the measurements will be compared with each other by considering their uncertainties. Finally, the average of all volume and mass measurements will be taken in each other and these values will be used to find their common densities. In this calculation, a relation between mass and volume is plotted with taking care of their errors. By analyzing this graph, the density of these objects will be calculated.

9.1. Density

Density, d, is an intrinsic property of materials. In fact, it is independent upon the amount of the object; therefore the density of anything remains the same, no matter the shape and size of the sample. The mass of an object may be distributed in a volume of an object uniformly (being homogeneous) or non-uniformly (being inhomogeneous). If the mass is distributed uniformly throughout the object, the density *d* is defined as the total mass m divided by the total volume *V* of the object. In equation form this is

$$d = \frac{m}{V}$$

9.2. Determination of External Dimensions



Figure-25: A set of objects with different shapes.

In this experiment, the dimensions of the objects will be measured. Therefore, we require the information about how to calculate the volume of the objects by using these measurements.

The **volume** of a rectangular solid is expressed with the equation of

$$\mathrm{V} = \mathrm{W} \times \mathrm{L} \times \mathrm{h}$$

$$\mathrm{V}[cm^3] = \mathrm{Area}[cm^2] \times \mathrm{height}[\mathrm{cm}]$$

where W, L and h are width, length and height, respectively.

Note that, the SI unit of kilogram per cubic meter (kg/m^3) and the CGS unit of gram per cubic centimeter (g/cm^3) are used units for density.

However, g/cm^3 is used commonly to define the density of an object. In general, scientists working in laboratories, dealing with small quantities and distances, prefer to measure distance in centimeters and mass in grams. Moreover, since the cubic centimeter can be alternatively converted to a milliliter or a cc, this fact may affect the usage of CGS system to define density.

SI unit system is based on meter-kilogram-second (MKS) unit set; however, their equivalents in CGS system unit are centimeter-gram-second.

In general, there is a common preference on the MKS system over the CGS system. Although some of the CGS units remain in use for a variety of purposes, their SI units equivalents in the MKS system are used as international standards.

For a sphere, the volume is given by

$$\mathrm{V} = \frac{4}{3}\pi \mathrm{r}^3$$

$$\mathrm{V}[cm^3] = \left(\frac{4}{3}\pi\right) \times \text{cubic radius}[cm^3]$$

where ***r*** is the radius of the sphere.

For a cylinder, the volume is given by

$$\mathrm{V} = \pi \mathrm{r}^2 \mathrm{L}$$

$$\mathrm{V}[cm^3] = \text{Area of the base circle}[cm^2] \times \text{length}[\mathrm{cm}]$$

where ***r*** is the cylinder radius, and ***L*** is its length.

A cone has a circular base and a vertex that is not on the base. Cones are similar in some ways to pyramids. They both have just one base and they converge to a point, the vertex. The formula for the volume of a cone is:

$$\mathrm{V} = \frac{1}{3}\pi \mathrm{r}^2 \mathrm{h}$$

$$\mathrm{V}[cm^3] = \left(\frac{1}{3}\right) \times \text{Area of the base circle}[cm^2] \times \text{height}[\mathrm{cm}]$$

where *r* is the radius of the base circle and *h* is height of the cone from base to the vertex.

Measuring the volume of the objects with ruler

There are two different scales on the ruler that is used in the experiment. One of them is that it has 0.5 mm divisions from 0 to 5 cm. The other is that it has 1 mm divisions from 0 to 15 cm.

If the dimension of the object is less than 5 cm, it is recommended to use first scale which is more sensitive than the other, if not use the main scale which has 1 mm divisions. When measuring the dimensions of the object, try to take the most possible readings considering the parallax. Then, record the result of the measurements for each object to the corresponding tables in the laboratory report.



Figure-26: Ruler.

For the circularly shaped objects, first measure the diameter and then convert it to radius.

Measurements in the cubes and rectangular solids are easier than the others since the ruler is holded parallel to the object.

Be careful to take the each measurement from the different regions of the object.

The height of the cylinder is also straightforward but the diameter of the base circles must be measured from its center. For the holes of the cylinder, ruler is not appropriate to measure its diameter and height (depth of the hole in the cylinder).

To take these measurements, you can benefit from using another object which can be easily inserted in the holes. The length of this object in the hole is equal to the height of the hole. For the cone, you can also use an object to determine its height by ruler. This straight object should be placed on the vertex and parallel to the base.

The height of the cone is equal to the distance between the base and the object on the ruler. For the sphere, again be careful while measuring its diameter as in the case of cylinder.

Measuring the volume of the objects with caliper



Figure-27: Caliper.

1. Loosen the locking screw and move the slider to check if the vernier scale works properly.
2. Check whether the surface of the external, internal and movable jaws, and also depth gauge are clean. If not, clean all these components.
3. Clean the measuring points (or surfaces) of the object which will be contact with the jaws during the measurement.
4. Before making any measurements, determine whether or not the vernier caliper reads zero when the jaws are fully closed. If the caliper does not read zero when the jaws are closed, they are said to have a zero error. Therefore, adjust the caliper's jaws until you get a zero reading. If it is not possible, you must add or subtract the offset from your readings.
5. For external measurements, open the jaws enough to place the **caliper** over the object (Figure-27).
6. For internal measurements, close the internal jaws and placed into object to be measured (**Figure-16**).
7. For the ***depth*** measurements, placed one end of the **caliper** against top of the hole (see Figure-17).
8. Slide lightly the movable jaw to make a contact between the surface of the external, internal or depth gauge with the surface of the object.
9. Tighten the locking screw so that the contact position will not be affected during measurement.
10. Read the main and Vernier scales.
11. Record your results to the corresponding tables in the Laboratory Report.

Measuring the volume of the objects with micrometer



Figure-28: Micrometer.

1. Loosen the lock nut and move the spindle to check if the vernier scale works properly.
2. Clean the anvil and spindle surfaces, and also the measuring object.
3. Before making any measurements, determine whether or not the micrometer reads zero when the anvil and spindle in contact. If the micrometer does not read zero when the anvil and spindle are fully closed, they are said to have a zero error.
4. Rotate the thimble enough to place the micrometer over the object.

5. Place the object in the micrometer so that it is in contact with the measuring surface of the object.

6. Rotate the thimble until the spindle is close to the other side of the object.

7. Turn the ratchet stop until you hear 3 clicks from the ratchet stop.

8. Verify that both the anvil and spindle are touching the object evenly.

9. Set the lock nut while the micrometer is still on the object.

10. Read the sleeve and thimble scales.

11. Record your results to the corresponding tables in the Laboratory Report.

Measuring the volume of the objects with graduated cylinder



Figure-29: Graduated cylinder.

1. Clean the graduated cylinder and the measuring object.

2. Fill the graduated cylinder up to a volume where the object can easily sink, and also this volume should be easily read on the scale of this container. Then, measure it as an initial volume.

3. Place the object in the container, taking care that the liquid level remains above that of the solid. Then, measure the final volume of the combined liquid and solid.

4. Record your results to the corresponding tables in the Laboratory Report.

Report each of the measurements as well as the results with their correct number of significant figures with their corresponding errors.

9.3. Determination of Mass

We can determine the mass of the objects with a laboratory balance by balancing the weight of an unknown mass against the weight of a known mass. Although the balance is between two forces (the weight of the masses), the scales can be calibrated in terms of mass assuming that the force per unit mass is the same for both the known and unknown mass. The unknown mass on a tray at one end of the beam is balanced against the sum of all the known masses placed on the other end. In our system, the slides on the beam will not be used in measurements. They will only be used to calibrate beam with zero reading on a scale. In addition, we can use digital balance as an alternative method to determine the mass of the objects. It will give a digital readout in grams.

Measuring the mass of the objects with equal arm balance



Figure-30: Equal arm balance.

1. Zero the balance before measuring the mass of any object.

   1.1. Be sure all riders are back to the **zero** point.

   1.2. Check to see that the pointer swings freely along the scale.

2. Place the object with an unknown mass in one tray and start adding mass (from a set of mass with different weights) to the other tray until the beam is balanced (parallel with the floor). The pointer must be in zero point if the arms are in equilibrium.

This mass set includes;

Table-**3**: Mass set.

Amount (piece)	Weight
1	500 g
1	200 g
2	100 g
1	50 g
1	20 g
2	10 g
1	5 g
2	2 g
2	1 g
1	500 mg
1	200 mg
1	100 mg
1	50 mg
1	20 mg
1	10 mg



Figure-31: A set of mass with different weights.

3. Record your results to the corresponding tables in the Laboratory Report.

Note that the smallest mass in the set is 10mg; therefore, the uncertainty in this instrument is taken as 10 mg.

Measuring the mass of the objects with digital balance

If you have any digital balance, it can be also used as an alternative method to determine the mass of the objects.



Figure-32: Digital balance.

1. Place the digital balance on a hard and smooth surface as to be parallel position with the floor.

2. Open the balance with pushing the ON/OFF button.

3. Select the appropriate unit of the measurement using MODE button. In addition to gram, there are 3 more measuring units; ounces (oz), pennyweight (dwt) and carat (ct).

1 oz = 28.350 g
1 dwt = 1.555 g
1 ct = 0.2 g

4. Zero the balance before measuring the mass of any object with pushing Z/T button.

5. Place the object with an unknown mass on a platform of the balance, and record the value on the screen.

Note that, the maximum limit of the balance used in this experiment is 200 g.

6. Record your results to the corresponding tables in the Laboratory Report.

While taking measurements, wait until the reading on the screen is stable.

Calculating the density of the objects

The material used to form the cube, rectangular solid, and cylinder and cone shaped objects are the same. In other words, their densities are equal to each other. Therefore, in order to determine this density value,

1. After completing the volume and mass measurements, take the average value of volume and mass for each object.

2. Using each pair of mass and volume values, plot a graph representing the relation between volume and mass.

   Note that, each mass and volume result from one of the objects will form a one data pair.

3. Construct this graph with mass plotted on the *y*-axis and volume plotted on the *x*-axis. Label the axes, and write down the corresponding unit on each axis. Plot the data points along with the error bars.

Note that the distribution of the data points suggests a straight line fitting. The equation of this line will have the familiar form $y = mx + b$, *where* ***m*** *represents the slope of the line, and* ***b*** *represents the* ***y****-intercept.*

4. Draw the best and worst straight lines for your data.

5. Find the slopes of the best line; m_b, and of the worst line, m_w. Calculate the error (uncertainty) in the slope $\Delta m = |m_b - m_w|$.

6. From these slopes, find the density of the unknown material as $d \pm \Delta d$.

The sphere shaped object has a different density as it is observed easily. For this object, again take the average of the results for mass and volume measurements. Then, calculate the density of this object by making proportion between these results.

9.4. Properties of Lenses

A lens is a transmissive optical device. It is typically a thin piece of glass or other transparent material suc as plastic that has the ability to redirect light rays to form an image. It is usually made in circular shape and its two surfaces are ground and polished in a specific usage. A simple lens consists of a single piece of material, while a compound lens consists of several simple lenses. In contrast to mirrors that reflects the incident light, and prisms which refract light without focusing, lenses can be used to focus light to form an image.

Lenses are designed to a desired shape so that to get a variety of types of lenses. Lenses differ from one another in terms of their shape and the materials from which they are made. The two most common types of lenses are concave and convex lenses in which incident rays can converge to a point or appear to be diverging from a point, respectively. These types of lenses are illustrated below in the **Figure-(33)**.

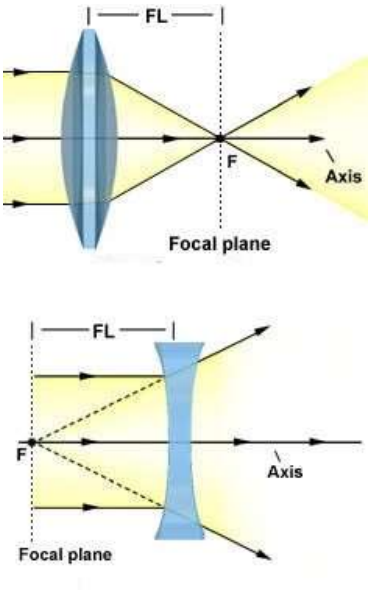


Figure-33: Action of (a) converging lens and (b) diverging lens on incident parallel light rays.

In a simple way, the point at which initially coming light rays meet after passing through a convex lens, or where the the extension of light rays refracted through the concave lens intersect is called a focal point of the lens.

Moreover, the distance between the center of the lens and focal point is defined as focal length. The horizontal line passing throug the center of the lens is called axis of the lens. The point on the surface of lens where the axis meets the lens is known as the vertex. The point in the center of the sphere from which the lens was sliced is known as the center of curvature. The distance from the vertex to the center of curvature is known as the radius of curvature.

Lenses are classified by the curvature of the two optical surfaces.A lens is *bi-convex* (or *double convex*, or just *convex*) if both surfaces are convex. If both surfaces have the same radius of curvature, the lens is *equiconvex*. A lens with two concave surfaces is *biconcave* (or just *concave*). If one of the surfaces is flat, the lens is *plano-convex* or *plano-concave* depending on the curvature of the other surface. A lens with one convex and one concave side is *convex-concave* or *meniscus*. It is this type of lens that is most commonly used in corrective lenses.



Figure-34: Bi-convex lens (a), plano-convex lens (b), concavo-convex lens (c), bi-concave lens (d), plano-concave lens (e) and convexo - concave lens (f).

The defining characteristic of a lens is a quantity called the focal length. Lenses with the same shape and index of refraction will have the same focal length. For different shaped lenses or for a lens in different medium, the focal length should be defined for proper use in optic systems. In other words, to determine whether a lens will behave as a converging or diverging lens based on the curvature of its faces and the relative indices of the lens material, n_1, and the surrounding medium, n_2, the focal length must be known.

For this purpose, the lens-maker's equation gives the reciprocal of the focal length in terms of the physical characteristics of the lens (and the medium in which the lens finds itself);

$$\frac{1}{f} = (n_2 - n_1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

where f is the focal length of the lens, n_1 is the index of refraction of the medium surrounding the lens, n_2 is the index of refraction of the material of which the lens is made, R_1 is the radius of curvature of one of the surface of the lens, and R_2 is the radius of curvature of the other surface of the lens. In the application of this equation, the important point is the algebraic sign convention for the R values. Since the convex lens has a *curved out* surfaces, by convention the radius of curvature for a convex surface is positive. On the other hand, with a *curved in* surfaces, in concave lens, by convention, the radius of curvature is negative.

10 Laboratory Report

Name and Surname: _____

Department: _____

Student Number: _____

Date: _____

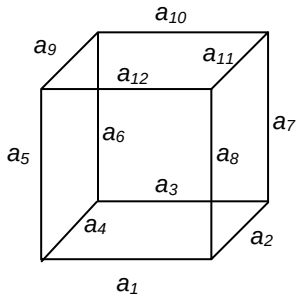
10.1. Measurements of the Object-1.

Data Table-1: Measuring dimensions of the object-1 by ruler (in cm).

Data Number	$a_1 \pm \Delta a_1$	$a_2 \pm \Delta a_2$	$a_3 \pm \Delta a_3$	$a_4 \pm \Delta a_4$	$a_5 \pm \Delta a_5$	$a_6 \pm \Delta a_6$	$a_7 \pm \Delta a_7$	$a_8 \pm \Delta a_8$	$a_9 \pm \Delta a_9$	$a_{10} \pm \Delta a_{10}$	$a_{11} \pm \Delta a_{11}$	$a_{12} \pm \Delta a_{12}$
1												
2												
3												
Average												

Data Table-2: Measuring dimensions of the object-1 by caliper (in mm).

Data Number	$a_1 \pm \Delta a_1$	$a_2 \pm \Delta a_2$	$a_3 \pm \Delta a_3$	$a_4 \pm \Delta a_4$	$a_5 \pm \Delta a_5$	$a_6 \pm \Delta a_6$	$a_7 \pm \Delta a_7$	$a_8 \pm \Delta a_8$	$a_9 \pm \Delta a_9$	$a_{10} \pm \Delta a_{10}$	$a_{11} \pm \Delta a_{11}$	$a_{12} \pm \Delta a_{12}$
1												
2												
3												
Average												



From the data in data Table-1, considering the object-1 to be a cube, take the final average of the all average values to determine dimension of "*a"* with its uncertainty. Then calculate $V \pm \Delta V$ and explicitly show the details of your calculation.

$a \pm \Delta a =$ ..... cm

$V \pm \Delta V =$ ..... cm^3

From the data in data Table-2, considering the object-1 to be a cube, take the final average of the all average values to determine dimension of "*a"* with its uncertainty. Then calculate $V \pm \Delta V$ and explicitly show the details of your calculation.

$a \pm \Delta a =$ ..... cm

$V \pm \Delta V =$ ..... cm^3

Data Table-3: Results of the calculation of the volume of the object-1 from dimensional measurements.

	By ruler	By caliper
$V \pm \Delta V$ (cm^3)		

Data Table-4: Measuring the volume of the object-1 by graduated cylinder.

Initial Volume (cm^3)	Final Volume (cm^3)	Volume Water Displaced (cm^3)	Volume of Object (cm^3)

Data Table-5: Measuring mass of the object-1.

	By Equal-Arm Balance (in unit of gram)	By digital balance (in unit of gram)
Data Number	$m \pm \Delta m$	$m \pm \Delta m$
1		
2		
3		
Average		

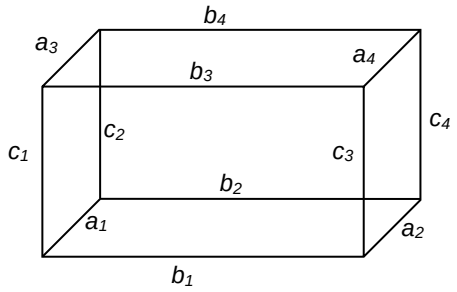
10.2. Measurements of the Object-2.

Data Table-6: Measuring dimensions of the object-2 by ruler (in cm).

Data Number	$a_1 \pm \Delta a_1$	$a_2 \pm \Delta a_2$	$a_3 \pm \Delta a_3$	$a_4 \pm \Delta a_4$	$b_1 \pm \Delta b_1$	$b_2 \pm \Delta b_2$	$b_3 \pm \Delta b_3$	$b_4 \pm \Delta b_4$	$c_1 \pm \Delta c_1$	$c_2 \pm \Delta c_2$	$c_3 \pm \Delta c_3$	$c_4 \pm \Delta c_4$
1												
2												
3												
Average												

Data Table-7: Measuring dimensions of the object-2 by caliper (in mm).

Data Number	$a_1 \pm \Delta a_1$	$a_2 \pm \Delta a_2$	$a_3 \pm \Delta a_3$	$a_4 \pm \Delta a_4$	$b_1 \pm \Delta b_1$	$b_2 \pm \Delta b_2$	$b_3 \pm \Delta b_3$	$b_4 \pm \Delta b_4$	$c_1 \pm \Delta c_1$	$c_2 \pm \Delta c_2$	$c_3 \pm \Delta c_3$	$c_4 \pm \Delta c_4$
1												
2												
3												
Average												



From the data in data Table-6, take the final average of the all average values to determine dimension of *a, b* and *c* with their uncertainties. Then calculate $V \pm \Delta V$ and explicitly show the details of your calculation.

$a \pm \Delta a =$..... cm

$b \pm \Delta b =$..... cm

$c \pm \Delta c =$..... cm

$V \pm \Delta V =$..... cm^3

From the data in data Table-7, take the final average of the all average values to determine dimension of *a, b* and *c* with their uncertainties. Then calculate $V \pm \Delta V$ and explicitly show the details of your calculation.

$a \pm \Delta a =$..... cm

$b \pm \Delta b =$..... cm

$c \pm \Delta c =$..... cm

$V \pm \Delta V =$..... cm^3

Data Table-8: Results of the calculation of the volume of the object-2 from dimensional measurements.

	by ruler	by caliper
$V \pm \Delta V$ (cm^3)		

Data Table-9: Measuring the volume of the object-2 by graduated cylinder.

Initial Volume (cm^3)	Final Volume (cm^3)	Volume Water Displaced (cm^3)	Volume of Object (cm^3)

Data Table-10: Measuring mass of the object-2.

	by equal arm balance (in unit of gram)	by digital balance (in unit of gram)
Data Number	$m \pm \Delta m$	$m \pm \Delta m$
1		
2		
3		
Average		

10.3. Measurements of the Object-3.

Data Table-11: Measuring dimensions of the object-3 by ruler (in cm).

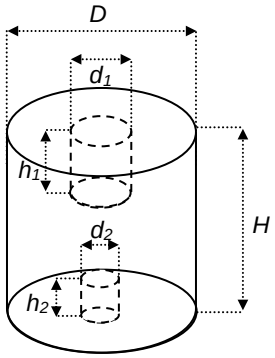
Data Number	$D \pm \Delta D$	$H \pm \Delta H$	$d_1 \pm \Delta d_1$	$h_1 \pm \Delta h_1$	$d_2 \pm \Delta d_2$	$h_2 \pm \Delta h_2$
1						
2						
3						
Average						

From the data in Table-11, determine dimension of *R*, *H*, r_1, h_1, r_2 and h_2 with their uncertainties. Then calculate $V \pm \Delta V$ of the object-3 and explicitly show the details of your calculation.

$V \pm \Delta V =$..... cm^3 (volume of the cylinder without the hole)

$V \pm \Delta V =$..... cm^3 (volume of the first hole)

$V \pm \Delta V =$..... cm^3 (volume of the second hole)



$V \pm \Delta V =$..... cm^3 (volume of the object-3)

Data Table-12: Measuring dimensions of the object-3 by caliper (in mm).

Data Number	$D \pm \Delta D$	$H \pm \Delta H$	$d_1 \pm \Delta d_1$	$h_1 \pm \Delta h_1$	$d_2 \pm \Delta d_2$	$h_2 \pm \Delta h_2$
1						
2						
3						
Average						

From the data in Table-12, determine dimension of *R, H*, r_1, h_1, r_2 and h_2 with their uncertainties. Then calculate $V \pm \Delta V$ of the object-3 and explicitly show the details of your calculation.

$V \pm \Delta V$ = ... mm^3 = ... cm^3 (volume of the cylinder without the hole)

$V \pm \Delta V$ = ... mm^3 = ... cm^3 (volume of the first hole)

$V \pm \Delta V$ =..... mm^3 = ... cm^3 ((volume of the second hole)

$V \pm \Delta V$ =..... mm^3 = ... cm^3 (volume of the object-3)

Data Table-13: Results of the calculation of the volume of the object-3 from dimensional measurements.

	by ruler	by caliper
$V \pm \Delta V (cm^3)$		

Data Table-14: Measuring the volume of the object-3 by graduated cylinder.

Initial Volume (cm^3)	Final Volume (cm^3)	Volume Water Displaced (cm^3)	Volume of Object (cm^3)

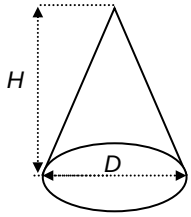
Data Table-15: Measuring mass of the object-3.

	By Equal-Arm Balance (in unit of gram)	By Digital Balance (in unit of gram)
Data Number		
1		
2		
3		
Average		

10.4. Measurements of the Object-4.

Data Table-16: Measuring dimensions of the object-4 by ruler (in cm).

Data Number	$D \pm \Delta D$	$H \pm \Delta H$
1		
2		
3		
Average		



From the data in Table-16, determine dimension of R and *H*, with their uncertainties.Then calculate $V \pm \Delta V$ of the object-4 and explicitly show the details of your calculation.

$V \pm \Delta V =$..... cm^3

Data Table-17: Measuring dimensions of the object-4 by caliper (in mm).

Data Number	$D \pm \Delta D$	$H \pm \Delta H$
1		
2		
3		
Average		

From the data in Table-17, take the final average of the all average values to determine dimension of R and *H* with their uncertainties. Then calculate $V \pm \Delta V$ of the object-4 and explicitly show the details of your calculation.

$V \pm \Delta V =$..... cm^3

Data Table-18: Results of the calculation of the volume of the object-4 from dimensional measurements.

	by ruler	by caliper
$V \pm \Delta V$ (cm^3)		

Data Table-19: Measuring the volume of the object-4 by graduated cylinder.

Initial Volume (cm^3)	Final Volume (cm^3)	Volume Water Displaced (cm^3)	Volume of Object (cm^3)

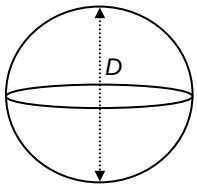
Data Table-20: Measuring mass of the object-4.

	By Equal Arm Balance (in unit of gram)	By Digital Balance (in unit of gram)
Data Number	$m \pm \Delta m$	$m \pm \Delta m$
1		
2		
3		
Average		

10.5. **Measurements of the Object-5.**

Data Table-21: Measuring dimensions of the object-5 by ruler, caliper and micrometer (in mm).

	by ruler	by caliper	by micrometer
Data Number	$D \pm \Delta D$	$D \pm \Delta D$	$D \pm \Delta D$
1			
2			
3			
Average			



From the data in Table-21, determine dimension of R, with its uncertainty for each measurements. Then calculate $V \pm \Delta V$ of the object-5 and explicitly show the details of your calculations.

$V \pm \Delta V =$..... mm^3 = ... cm^3 (by ruler)

$V \pm \Delta V =$..... mm^3 = ... cm^3 (by caliper)

$V \pm \Delta V =$..... mm^3 = ... cm^3 (by micrometer)

Data Table-22: Results of the calculation of the volume of the object-5 from dimensional measurements.

	by ruler	by caliper	by micrometer
$V \pm \Delta V$ (cm^3)			

Data Table-23: Measuring the volume of the object-5 by graduated cylinder.

Initial Volume (cm^3)	Final Volume (cm^3)	Volume Water Displaced (cm^3)	Volume of Object (cm^3)

Data Table-24: Measuring mass of the object-5.

	by Equal Arm Balance (in unit of gram)	by Digital Balance (in unit of gram)
Data Number	$m \pm \Delta m$	$m \pm \Delta m$
1		
2		
3		
Average		

Determine the **density** of the object-5 from the measured data in **Table-22** and **Table-24** (Show the details of the calculation explicitly).

10.6. Density Calculation

Data Table-25: Summary of the volume measurements for the objects 1-4.

	Volume (cm^3) by Ruler	Volume (cm^3) by Caliper	Volume (cm^3) by Graduated Cylinder	Average value of Volume (cm^3)
Object-1				
Object-2				
Object-3				
Object-4				

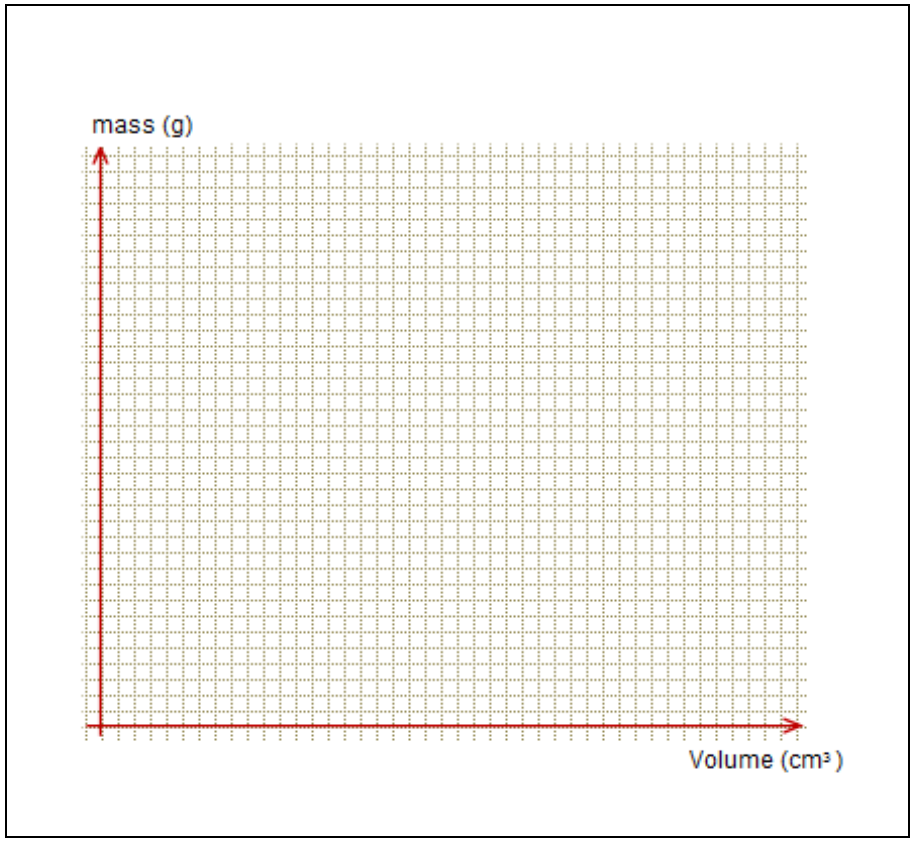
Data Table-26: Summary of the mass measurements for the objects 1-4.

	Mass (g) by Equal Arm Balance	Mass (g) Digital Balance	Average value of Mass (g)
Object-1			
Object-2			
Object-3			
Object-4			

We know that tall these four objects are made from the same material. Then, taking the average volume and mass data set from each independent measurements, plot the graph of the average volume and mass of the objects. For simplicity, put the volume values at the vertical axis, and the mass values at the horizontal axis. Draw the data points along with the corresponding error bars. Noting that the distribution of the data points on the graph paper suggests a straight line fitting, draw the best and worst straight lines. Find the slope $\boldsymbol{m}$ of the best line and the corresponding error $\boldsymbol{\Delta m}$. What are the units of this slope?. What physical quantity does the slope represent?.

$m_b =$ . . . . . .

$m_w =$ . . . . . .



Graph-1: The change in the mass of the objects with respect to the volume.

Report $m \pm \Delta m$ with the correct number of significant figures and units,

$m \pm \Delta m =$ . . . . . .